

MATH9102 Fundamentals of Data Analysis

TU059/TU060/TU256/Erasmus

CA Phase ONE Marking Scheme/Rubric

Due Date: Friday 7th November 2025 @ 23:59 (via Brightspace)

Contributes 40% to overall module marks

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Marking Breakdown

ASPECT	ISSUES CONSIDERED		OVERALL MARKS
HYPOTHESES	- Clarity and appropriateness of four hypotheses. - Correct identification of variable types involved.	4 x 2 marks 4 x 1 marks	12 Marks
DATA EXPLORATION	- Dataset general description. - Data suitability inspection. - Descriptive statistics and visualisations appropriate to hypotheses.	2 marks 8 marks 4 x 2 marks	18 marks
ANALYSIS AND TEST SELECTION	- Correct choice of statistical tests for each hypothesis. - Evidence-based justification for choices derived from data exploration.	4 x 2 marks 4 x 4 marks	24 marks
EXECUTION AND INTERPRETATION	- Correct execution of statistical tests in R. - Correct interpretation of statistical tests. - Accurate reporting of results (tables, graphs, statistical language).	4 x 1 marks 4 x 3 marks 4 x 3 marks	28 marks
CONCLUSIONS	Clear conclusions for each hypothesis, linked back to evidence.	4 x 2 marks	8 marks
CODING	Organisation, clarity, comments, reproducibility in Quarto.		5 marks
DOCUMENTATION	Report structure, layout, clarity, grammar, adherence to submission requirements.		5 marks
Total Marks Available ¹			100 marks

¹ The task will be marked out of 100 and your result will be weighted to reflect the 40% of in the calculation of the overall module marks.

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Overall Levels of Achievement

ACHIEVEMENT	EXCELLENT	GOOD	SATISFACTORY		UNSATISFACTORY
% OF MARKS AVAILABLE	> 75%	60-74%	40-59%		<40%
Key Characteristics associated with each achievement level for each aspect are described on the next pages. ²					

² The characteristics described below in the rubric are for a student submission which addresses FOUR(4) hypotheses. Where a student attempts fewer than FOUR(4), the marks achieved will depend on how many hypotheses have been addressed and the level of achievement for each question. For example, if a student attempts only THREE and completes everything to the level indicated by the category of EXCELLENT, their mark will not reach the level associated with this category but will be that associated with a lower category.

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HYPOTHESES Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
- Four hypotheses stated clearly and precisely. - Correctly identifies variable types. - Hypotheses are testable and relevant to dataset context.	- Four hypotheses included with minor errors/ambiguity. - Variable types mostly identified correctly. - Hypotheses testable but lacking refinement.	- Some hypotheses vague or inappropriate. - Errors in identifying variable types. - Hypotheses weakly linked to dataset.	- Hypotheses missing, incomplete, or largely incorrect. - Variable types not identified or misclassified.

DATA EXPLORATION Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
- Dataset general description is clear and complete (source, size, variables, collection context). - Dataset suitability inspection is thorough and justified. - Variables described clearly, with relevant descriptive statistics. - Appropriate and well-presented visualisations support analysis.	- Dataset general description included but with minor omissions. - Dataset inspection mostly clear with minor gaps. - Statistics and visualisations appropriate but limited. - Reasonable demonstration of dataset suitability.	- Dataset general description is brief or incomplete. - Dataset inspection attempted but limited. - Statistics/visualisations included but with errors or omissions. - Weak justification of dataset suitability.	- No meaningful dataset description. - No meaningful inspection of dataset. - Inappropriate/missing statistics and visualisations. - No evidence dataset is suitable for analysis.

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ANALYSIS AND TEST SELECTION Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
• Correct statistical tests selected for all hypotheses. • Justification fully aligned with data inspection. • Evidence-based reasoning clearly articulated.	• Tests appropriate with minor errors or weaker justification. • Reasoning mostly aligned with data inspection.	• Some tests correct, others inappropriate. • Justification vague or partially inconsistent with exploration.	• Tests missing or incorrect. • No meaningful justification.

EXECUTION AND INTERPRETATION Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
• Tests executed correctly in R. • Results reported accurately with clear tables, graphs, and statistical notation. • Reporting uses correct terminology.	• Correct tests executed with minor errors. • Results reported clearly with some omissions or small mistakes.	• Execution or reporting contains significant errors. • Results incomplete or unclear.	• Tests not executed or incorrectly executed. • Results absent or meaningless.

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CONCLUSIONS Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
• Clear, concise conclusions for each hypothesis. • Conclusions directly supported by test results and evidence.	• Conclusions included but less precise. • Links to results present but minor gaps.	• Conclusions vague, incomplete, or only loosely linked to evidence.	• Conclusions absent or not linked to evidence.

CODING Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
• Code clearly organised and labelled in Quarto. • Helpful comments included. • Code runs without adaptation.	• Code clear with most major steps commented. • Code runs without adaptation.	• Code somewhat difficult to follow. • Minimal comments. • Requires effort to reproduce.	• Code disorganised or difficult to follow with minimal comments/ Code incomplete or non-functional. • Requires significant effort to reproduce.

DOCUMENTATION Rubric

EXCELLENT	GOOD	SATISFACTORY	UNSATISFACTORY
• Report well-structured, polished, free from grammatical errors. • Submission fully adheres to conventions required.	• Report structured and clear, with minor grammar/spelling issues. • Minor deviations from conventions required.	• Report somewhat disorganised, with several grammar/spelling errors. • Deviates from conventions required.	• Report poorly organised, unclear, with major grammar/spelling issues. • Does not adhere to conventions required.